# Business Question and Visualization Report - Snowflake

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| **Date** | 29 October 2026 |
| **Team ID** | Individual Project - Ujjawal Rathod |
| **Project Name** | Electric Vehicle Market Performance & Insights |
| **Maximum Marks** | 5 Marks |

# Purpose

This report connects EV market business questions to Snowflake SQL outputs and recommended visualizations. The same questions support the Tableau dashboard story and the three stakeholder scenarios: market strategy, competitive intelligence, and sustainable mobility.

*SQL examples use a canonical curated schema. Map the source CSV headers to these aliases in the staging view before running: MODEL_YEAR, MARKET_REGION, VEHICLE_CATEGORY, MANUFACTURER, BATTERY_TYPE, DRIVING_RANGE_KM, ENERGY_CONSUMPTION_WH_KM, CUSTOMER_RATING, MANUFACTURING_COUNTRY, SALES_VOLUME. The supplied handoff did not include the raw CSV files or a live Snowflake connection, so numeric SQL outputs are intentionally not invented.*

*Evidence note: the live Snowflake query outputs and screenshots were not supplied. Each item lists the output fields and screenshot evidence to capture after running the query. Do not treat a design description as a live execution result.*

## 1. Which vehicle categories contribute the highest sales volume?

### Snowflake SQL

```
SELECT VEHICLE_CATEGORY, SUM(SALES_VOLUME) AS TOTAL_SALES_VOLUME
FROM EV_CURATED
GROUP BY VEHICLE_CATEGORY
ORDER BY TOTAL_SALES_VOLUME DESC;
```

### Recommended visualization and business use

Horizontal bar chart; category on rows, total sales volume on columns, descending sort. Compare category mix before allocating product or marketing effort.

| Output fields | Evidence / validation |
|---|---|
| VEHICLE_CATEGORY, TOTAL_SALES_VOLUME | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 2. How has EV sales volume changed across model years?

## Snowflake SQL

```
SELECT MODEL_YEAR, SUM(SALES_VOLUME) AS TOTAL_SALES_VOLUME
FROM EV_CURATED
GROUP BY MODEL_YEAR
ORDER BY MODEL_YEAR;
```

## Recommended visualization and business use

Line chart; model year on the x-axis and total sales volume on the y-axis. Use the 2018-2026 range consistently.

| Output fields | Evidence / validation |
|---|---|
| MODEL_YEAR, TOTAL_SALES_VOLUME | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 3. Which manufacturers lead by sales, and how fragmented is market share?

## Snowflake SQL

```
WITH totals AS (
  SELECT MANUFACTURER, SUM(SALES_VOLUME) AS MANUFACTURER_SALES
  FROM EV_CURATED GROUP BY MANUFACTURER
), market AS (
  SELECT SUM(MANUFACTURER_SALES) AS MARKET_SALES FROM totals
)
SELECT t.MANUFACTURER, t.MANUFACTURER_SALES,
      100.0 * t.MANUFACTURER_SALES / NULLIF(m.MARKET_SALES, 0) AS MARKET_SHARE_PCT
FROM totals t CROSS JOIN market m
ORDER BY t.MANUFACTURER_SALES DESC
LIMIT 10;
```

## Recommended visualization and business use

Ranked horizontal bars plus percent labels. Pair the Top 10 view with a fixed all-market denominator so a filtered view does not misleadingly total 100%.

| Output fields | Evidence / validation |
|---|---|
| MANUFACTURER, MANUFACTURER_SALES, MARKET_SHARE_PCT | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 4. What is the adoption mix of battery technologies?

## Snowflake SQL

```
SELECT BATTERY_TYPE, SUM(SALES_VOLUME) AS SALES_VOLUME,
    100.0 * SUM(SALES_VOLUME) / NULLIF(SUM(SUM(SALES_VOLUME)) OVER (), 0) AS SHARE_PCT
FROM EV_CURATED
GROUP BY BATTERY_TYPE
ORDER BY SALES_VOLUME DESC;
```

## Recommended visualization and business use

Donut chart for share and a sorted bar chart for comparison. Keep all battery categories visible and label the percentage.

| Output fields | Evidence / validation |
|---|---|
| BATTERY_TYPE, SALES_VOLUME, SHARE_PCT | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 5. How does average driving range differ by vehicle category?

## Snowflake SQL

```
SELECT VEHICLE_CATEGORY, AVG(DRIVING_RANGE_KM) AS AVG_RANGE_KM
FROM EV_CURATED
WHERE DRIVING_RANGE_KM IS NOT NULL
GROUP BY VEHICLE_CATEGORY
ORDER BY AVG_RANGE_KM DESC;
```

## Recommended visualization and business use

Sorted bar chart; average range in km by category. The earlier project summary reported a range from about 105 km for scooters to about 550 km for luxury sedans; verify these in query output.

| Output fields | Evidence / validation |
|---|---|
| VEHICLE_CATEGORY, AVG_RANGE_KM | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 6. How do battery technologies compare on range and energy consumption?

## Snowflake SQL

```
SELECT BATTERY_TYPE,
    AVG(DRIVING_RANGE_KM) AS AVG_RANGE_KM,
    AVG(ENERGY_CONSUMPTION_WH_KM) AS AVG_ENERGY_WH_KM
FROM EV_CURATED
GROUP BY BATTERY_TYPE
ORDER BY AVG_RANGE_KM DESC;
```

## Recommended visualization and business use

Scatter plot with average energy consumption on x and average range on y; color by battery type. Interpret range and efficiency together, not range alone.

| Output fields | Evidence / validation |
|---|---|
| BATTERY_TYPE, AVG_RANGE_KM, AVG_ENERGY_WH_KM | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 7. How does customer satisfaction vary by vehicle category?

## Snowflake SQL

```
SELECT VEHICLE_CATEGORY, AVG(CUSTOMER_RATING) AS AVG_CUSTOMER_RATING,
    COUNT(CUSTOMER_RATING) AS RATED_RECORDS
FROM EV_CURATED
WHERE CUSTOMER_RATING BETWEEN 1 AND 5
GROUP BY VEHICLE_CATEGORY
ORDER BY AVG_CUSTOMER_RATING DESC;
```

## Recommended visualization and business use

Bar chart with rating axis fixed to 0-5 and record count in tooltip. Earlier handoff described average ratings around 4.08-4.09; recheck using the actual query.

| Output fields | Evidence / validation |
|---|---|
| VEHICLE_CATEGORY, AVG_CUSTOMER_RATING, RATED_RECORDS | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 8. Which manufacturing countries contribute most, and how do market regions compare?

## Snowflake SQL

```
SELECT MANUFACTURING_COUNTRY, MARKET_REGION, SUM(SALES_VOLUME) AS SALES_VOLUME
FROM EV_CURATED
GROUP BY MANUFACTURING_COUNTRY, MARKET_REGION
ORDER BY SALES_VOLUME DESC;
```

## Recommended visualization and business use

Heatmap: country by region, colored by sales volume; add country and region filters. Avoid assuming manufacturing country is the same as sales destination.

| Output fields | Evidence / validation |
|---|---|
| MANUFACTURING_COUNTRY, MARKET_REGION, SALES_VOLUME | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# 9. How evenly is sales volume distributed across the market regions?

## Snowflake SQL

```
SELECT MARKET_REGION, SUM(SALES_VOLUME) AS TOTAL_SALES_VOLUME,
    100.0 * SUM(SALES_VOLUME) / NULLIF(SUM(SUM(SALES_VOLUME)) OVER (), 0) AS SHARE_PCT
FROM EV_CURATED
GROUP BY MARKET_REGION
ORDER BY TOTAL_SALES_VOLUME DESC;
```

## Recommended visualization and business use

Sorted regional bar chart with percentage labels, supported by a map if region names have a reliable geographic mapping.

| Output fields | Evidence / validation |
|---|---|
| MARKET_REGION, TOTAL_SALES_VOLUME, SHARE_PCT | Capture the executed SQL result grid and the rendered chart from the live environment. Check that totals reconcile to the curated dataset. |

# Submission checklist

- Run each query against the verified curated view and save the result grid.
- Insert actual screenshots for the code output and visualization in the final submission copy.
- Record the exact schema/database/view names used in Snowflake.
- Validate all metrics against the Tableau workbook before presenting stakeholder conclusions.